IEORE4004: Optimization Models & Methods **Fall 2025**

Zero-Sum games

1 Paper-Scissor-Rock as a Zero-Sum game

1.1 Introduction

Consider the classical Paper-Scissor-Rock (PSR) game, where two players play simultaneously one of P,S, and R:

- If they play the same, they draw;
- If one plays R, and the other play S, she who plays R wins;
- If one plays S, and the other play P, she who plays S wins;
- If one plays P, and the other play R, she who plays P wins.

If we think of each player winning (or losing) 1 (say, dollar), PSR is an example of **Zero-Sum** game: whatever one player wins, the other loses. We can represent the possible outcomes of the game by a **payoff matrix** A, where each row corresponds to a possible play of the first player (that from now on we call row player – rp), and each column to a play of the second player (that from now on we call column player – cp).

$$A = \begin{array}{c|ccc} & P & S & R \\ \hline P & 0 & 1 & -1 \\ S & -1 & 0 & 1 \\ R & 1 & -1 & 0 \end{array}$$

Hence, A_{ij} is the gain of cp when she plays j and rp plays i; equivalently (since the game is Zero-sum), it is the loss of rp when she plays i and cp plays j (a loss of -1 is a gain of 1 and similarly a gain of -1 is a loss of 1).

A **pure strategy** for a player is a choice of which of P, S, R to play. For instance, cp could choose to play R. That wouldn't be a good choice, since if rp knows her strategy, she would play P, and win. So what makes the problem more interesting are **mixed** (or **randomized**) strategies, where each player chooses a certain play with a given probability. Now the question becomes more interesting: which is the best strategy of each player? We will show LP and duality can help to answer this question. **In the following, we assume the game under consideration is PSR above, but the arguments easily extend to any payoff matrix A.**

1.2 Computing the strategies

As mentioned above, a strategy for cp is a 3-dimensional vector $x = (x_P, x_S, x_R)^T$, where x_i is the probability that cp plays i, for $i \in \{P, R, S\}$. Hence, x satisfies

$$x_P + x_R + x_S = 1, \quad x_P, x_R, x_S \geq 0.$$

Similarly, a strategy for rp is a 3-dimensional vector $y = (y_P, y_S, y_R)$ satisfying

$$y_P + y_R + y_S = 1, \quad y_P, y_R, y_S \geq 0. \tag{1}$$

If the strategy of cp is $\bar{x}$ and the strategy of rp is $\bar{y}$, the expected payoff for cp is

$$\sum_i \sum_j \bar{y}_i \bar{x}_j A_{ij} = \bar{y}^T A \bar{x} \tag{2}$$

and the expected payoff for rp is minus this number.

Suppose now that cp decides to play strategy $\bar{x}$, and rp knows about this. Which should be rp's strategy? She clearly want to choose a feasibly strategy (i.e., satisfying (1)) that maximizes the expected payoff (2). That is, she wants to solve

$$\begin{array}{lrcl} \min & y^T A \bar{x} & & \\ & y_P + y_R + y_S & = & 1 \\ & y_P, y_R, y_S & \geq & 0 \end{array}$$

(Note that $\bar{x}$ here is given – that's the strategy of cp, which we assume fixed – while y are the decision variables). Since $A\bar{x}$ is a vector, say q, rp should just pick the component of q of minimum value, say component i, and set $\bar{y}_i = 1$, $\bar{y}_j = 0$ for $j \neq i$. That is, rp wants to solve

$$\min_{i \in \{R,P,S\}} [A\bar{x}]_i. \tag{3}$$

(Here $[A\bar{x}]_i$ is the i-th entry of $A\bar{x}$). So cp can assume that, whatever strategy $\bar{x}$ she chooses, rp will solve (3) to find her optimal strategy. Hence, cp wants to choose $\bar{x}$ so that the solution of (3) is as large as possible. That is, cp wants to solve

$$\begin{array}{ll} \max_x & \min_{i \in \{R,P,S\}} [Ax]_i \\ & x_P + x_R + x_S = 1 \\ & x_P, x_R, x_S \geq 0 \end{array}$$

This seems a terribly non-linear problem, but it's actually not. By using ideas similar to the "linear version" of linear regression, one can show it is equivalent to the following problem, where we added one variable v:

$$\begin{array}{lrcl} \max v & & & \\ & [Ax]_R & \geq & v \\ & [Ax]_S & \geq & v \\ & [Ax]_P & \geq & v \\ & x_P + x_R + x_S & = & 1 \\ & x_P, x_R, x_S & \geq & 0 \end{array} \tag{4}$$

So cp will solve the problem above as to obtain the optimal solution $(\bar{x}, \bar{v})$, and use $\bar{x}$ as her optimal strategy. What about rp? It can employ very similar arguments, and find her optimal strategy $\bar{y}$ by solving the problem

$$\begin{array}{ll} \min_y & \max_{i \in \{R,P,S\}} [Ax]_i \\ & y_P + y_R + y_S = 1 \\ & y_P, y_R, y_S \geq 0 \end{array}$$

which is equivalent to

$$
\begin{aligned}
&\min u \\
&\qquad \begin{array}{lcl}
[y^T A]_R & \leq & u \\
[y^T A]_S & \leq & u \\
[y^T A]_P & \leq & u \\
y_P + y_R + y_S & = & 1 \\
y_P, y_R, y_S & \geq & 0
\end{array}
\end{aligned}
\tag{5}
$$

Now, let's argue that (4) and (5) are dual to each other. (4) can be written as:

$$
\begin{array}{llcl}
\max\, [1, \mathbf{0}^T] \begin{bmatrix} v \\ x \end{bmatrix} & & & \\
\text{s.t.} & (\mathbf{1},\ -A) \begin{bmatrix} v \\ x \end{bmatrix} & \leq & \mathbf{0} \\
& (0,\ \mathbf{1}^T) \begin{bmatrix} v \\ x \end{bmatrix} & = & 1 \\
& x & \geq & 0.
\end{array}
$$

where $\mathbf{1}$ (resp. $\mathbf{0}$) is the column vector of all ones (resp. zeros) that has as many components as x. Similarly (5) can be written as:

$$
\begin{array}{llcl}
\min\, [1, \mathbf{0}^T] \begin{bmatrix} u \\ y \end{bmatrix} & & & \\
\text{s.t.} & (\mathbf{1},\ -A^T) \begin{bmatrix} u \\ y \end{bmatrix} & \geq & \mathbf{0} \\
& (0,\ \mathbf{1}^T) \begin{bmatrix} u \\ y \end{bmatrix} & = & 1 \\
& y & \geq & 0.
\end{array}
$$

Using standard theory one can check that they are dual to each other (do that as an exercise). Hence, by LP duality theory, their optimal solutions have the same objective function value. That is:

$$\max_x \min_y y^T A x = \min_y \max_x y^T A x$$

and the strategy of the players will converge to the solution achieving this payoff (which in the case of the PSR game is, by symmetry, 0).

1.3 Explicit computation of the strategies

Let us now write the linear program (4) more explicitly.

$$
\begin{aligned}
&\max v \\
&\qquad \begin{array}{lcl}
x_P - x_S & \geq & v \\
-x_P + x_R & \geq & v \\
x_S - x_R & \geq & v \\
x_P + x_R + x_S & = & 1 \\
x_P, x_R, x_S & \geq & 0
\end{array}
\end{aligned}
\tag{6}
$$

Similarly, we can rewrite (5) as

$$\begin{aligned} \min u \quad & \\ -y_P + y_S &\le u \\ y_P - y_R &\le u \\ -y_S + y_R &\le u \\ y_P + y_R + y_S &= 1 \\ y_P, y_R, y_S &\ge 0 \end{aligned} \tag{7}$$

We claim that the optimal solution to (6) and (7) are $\begin{pmatrix} x^* \\ 0 \end{pmatrix} = \begin{pmatrix} y^* \\ 0 \end{pmatrix} = \begin{pmatrix} 1/3 \\ 1/3 \\ 1/3 \\ 0 \end{pmatrix}$. We can indeed verify that they are feasible for both problem, and they both have objective function equal to 0. Because (7) is equivalent to the dual of (6), by strong duality we conclude that they are the optimal solutions to these problems. So for both players, the optimal strategy is to play each of P,R,S with probability 1/3.

2 Another example

Consider the following 3×3 zero-sum game whose entries are *cp*'s payoffs (so rp's payoffs are their negatives):

$$A = \begin{pmatrix} 1 & 0 & -2 \\ 3 & -1 & -1 \\ -2 & 1 & 1 \end{pmatrix}.$$

Repeating the arguments made for the PSR game, we can obtain the optimal strategy for the column player by solving the following linear program.

$$\begin{array}{lll} \max & v & \\ \text{s.t.} & x_1 + 0\,x_2 - 2x_3 \ge v & \text{(row 1)} \\ & 3x_1 - x_2 - x_3 \ge v & \text{(row 2)} \\ & -2x_1 + x_2 + x_3 \ge v & \text{(row 3)} \\ & x_1 + x_2 + x_3 = 1, & \\ & x_1, x_2, x_3 \ge 0. & \end{array} \tag{8}$$

Similarly, the optimal strategy for the row player is given by the following LP

$$\begin{array}{lll} \min & u & \\ \text{s.t.} & 1\,y_1 + 3y_2 - 2y_3 \le u & \text{(column 1)} \\ & 0 \cdot y_1 - y_2 + y_3 \le u & \text{(column 2)} \\ & -2y_1 - y_2 + y_3 \le u & \text{(column 3)} \\ & y_1 + y_2 + y_3 = 1, & \\ & y_1, y_2, y_3 \ge 0. & \end{array} \tag{9}$$

Solution. To compute the optimal strategies, this time we cannot resort to symmetry, and we have instead to use a solver. An optimal strategy for rp is

$$\bar{y}^T = (0, \tfrac{3}{7}, \tfrac{4}{7}),$$

and an optimal strategy for cp can be chosen as

$$\bar{x}^T = \left(\tfrac{2}{7}, \tfrac{5}{7}, 0\right)$$

The value of the game (cp's expected payoff) is

$$\bar{v} = \tfrac{1}{7}$$

Equivalently, rp's expected payoff is $-\frac{1}{7}$.

We can check this is correct by plugging $\binom{\bar{x}}{v}$ into (8) and $\binom{\bar{y}}{v}$ into (9).